

PAPER_1674_SIGMA_LBL_LAMBDA_POW_4

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PAPER_1674 — Light-by-Light Cross Section = σ_{LbL} = $\alpha \blacksquare = \Lambda \blacksquare$ EXACT identity

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: QED
Date: June 18, 2026
Status: CLOSED — EXACT formula closure (PAPER_136x broader corpus)

Observation

PAPER_1374 (PAPER_136x broader corpus) gives:

``
 $\sigma_{\text{LbL}} = \alpha \blacksquare = \Lambda \blacksquare$ EXACT identity
``

Status: **EXACT formula.**

UQFF Closed Identity

``
 $\sigma_{\text{LbL}} = \alpha \blacksquare = \Lambda \blacksquare$ EXACT identity EXACT formula
``

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Empirical qed measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1374
- Related: PAPER_1656-1665 (tier-24); PAPER_1646-1655 (tier-23); PAPER_1636-1645 (tier-22)
- Calculator dispatch: `calculate_paradox({"paradox": "sigma_lbl_lambda_pow_4"})`

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